Mid-Review Q&A Preparation

Robust Reconstruction of Tokamak Plasma State from Sparse Diagnostics Using Continuous-Time Neural Networks

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Q1. What exactly are you reconstructing, and from what?

A: We reconstruct the 2D poloidal magnetic flux map $\psi(R, Z)$ (TokaMark Task 1-3) – and, as a simpler secondary check, the plasma boundary contour (Task 1-2) – from a short window of sparse magnetic sensor readings: flux loops, pickup coils, and poloidal-field coil currents. This is the same task definition and the same real data (FAIR-MAST, real MAST tokamak shots) used by the TokaMark benchmark, so our results are directly comparable to their published baseline.

Q2. Why did you move away from a synthetic simulation to real data?

A: Our original plan used a synthetic plasma simulation as a stand-in for fusion data. On checking the closest prior-art paper more carefully, we found its simulation testbed was actually built for a Hall-thruster (electric propulsion) device, not a magnetic-confinement fusion reactor – a different physical regime entirely. We replaced it with TokaMark, a real, public benchmark built on real MAST tokamak shots, which also came with an existing published baseline (CNN and CNN+LSTM) we can directly reproduce and compare against.

Q3. Is plasma reconstruction itself a solved problem?

A: Under clean, in-distribution conditions – yes, reasonably well. TokaMark's own CNN+LSTM baseline gets 0.159 NRMSE on Task 1-3 under a random train/test split (v3, Table 3: CNN+LSTM 0.1591; CNN 0.1481), and a transformer foundation model (TokaMind, 2026) does even better. What is comparatively less studied: performance under distribution shift (TokaMark's own temporal split raises the error to 0.257 for Task 1-3, v3 Table 3: CNN+LSTM 0.2567; CNN 0.5016) and performance under sensor corruption at test time, which – to our knowledge – has not yet been measured for this task.

Q4. What exactly is the research gap? Is it really unexplored, or has someone already done this?

A: We checked this thoroughly, including searches specifically designed to find a counterexample. Sensor-corruption robustness *has* been rigorously tested on real tokamak data – but only for one forecasting task (Gupta, 2026, predicting future plasma current); it is comparatively less studied for a spatial reconstruction task. Continuous-time models *have* been applied to tokamak data – but only for modeling plasma transport dynamics or scalar forecasting; their application to reconstructing a spatial field from sensors appears underexplored. The specific combination – continuous-time encoder + reconstruction task + measured robustness to sensor corruption – was not found in any paper we reviewed. We state this as an **underexplored evaluation**, a limited-work claim, not an absolute "no one has ever thought of this" claim.

Q5. Why does robustness matter here specifically?

A: Real magnetic diagnostics degrade, drop out, or fail during operation, and next-generation reactors face additional neutron-induced degradation of in-vessel magnetics over their operating life. A reconstruction model that only works when every sensor is perfect is of limited practical use for real-time control. Gupta's benchmark already showed this failure mode is severe for a related task (LSTM's forecasting error rose by 212% under sensor failure); we are checking whether the same severity occurs for reconstruction, and whether a different architecture handles it better.

Q6. What exactly did Gupta (2026) do, and what didn't they do?

A: Gupta trained four separate models – LSTM, Transformer, CNN, and XGBoost – normally, on clean data, for a forecasting task (predicting plasma current 100 ms ahead). He then froze each trained model and fed it deliberately corrupted test data under six scenarios (random dropout, single-sensor failure, temporal gaps, and others), measuring how much each model's error increased. He also tested three data-level fixes (zero-fill, mean-fill, forward-fill) applied after corruption. He did **not** test a combined CNN+LSTM architecture (TokaMark's own baseline uses that combination; Gupta's models were separate), and he did **not** test any reconstruction task. Note also that this is a solo-authored preprint by an independent researcher, not yet peer-reviewed – credible and reproducible (code and data are public), but not an established, heavily-cited result.

Q7. Why continuous-time models specifically, rather than just improving the data (imputation, masking)?

A: Every existing fix for missing/corrupted sensor data in this domain is data-level: Gupta's imputation strategies, and TokaMind's masked-dropout training. An *architectural* fix – a different model class handling the problem by design – is comparatively less explored. Continuous-time models (CfC, LTC) take the actual elapsed time between readings as an explicit input, so a longer-than-usual gap changes the model's internal state update rather than being silently treated the same as a normal step, as an LSTM would. This is a theoretical reason to expect better degradation behavior – our project tests whether that theoretical advantage actually holds.

Q8. Has anyone applied CfC, LTC, or Neural ODEs to tokamak reconstruction before?

A: Neural ODEs have been applied to tokamak research, but only for modeling plasma transport dynamics (energy exchange between plasma regions) and for forecasting scalar quantities like plasma inductance – not for reconstructing a spatial field from sensors. A related but architecturally different continuous-time model (a "Continuous CNN," built from continuous convolution kernels rather than a recurrent continuous-time state) was applied to disruption prediction and significantly outperformed a discrete CNN (AUC 0.974 vs. 0.799) – but that paper explicitly states they have not yet tested robustness to missing data, and it is a classification task, not reconstruction. In the literature we reviewed, we found no application of CfC or LTC specifically to any tokamak task, reconstruction or otherwise – this appears to be limited work, though we cannot claim to have surveyed every paper in the field.

Q9. What are your sensor corruption scenarios, exactly?

A: Adapting Gupta's methodology from forecasting to reconstruction, applied to the test set only (models are always trained on clean data):

1. **Random dropout** – randomly zero a fraction (10% / 25% / 50%) of sensor readings across

channels and timesteps

2. **Single-channel failure** – one full sensor (e.g. a specific flux loop) removed for the entire input window

3. **Temporal gap** – a contiguous block of timesteps blanked out, simulating an acquisition dropout

Q10. What are your baselines and evaluation metrics?

A: Baselines: TokaMark v3's own published CNN and CNN+LSTM models for Task 1-3/1-2, which we reproduce first as E1 (correctness check, not a claim) against v3 Table 3 (0.148/0.159 NRMSE, random split). **Metric**: NRMSE, matching TokaMark's own metric for direct comparability, plus a robustness score expressing relative degradation from clean to corrupted conditions, following Gupta's framing. **Design**: E2 (main) compares CNN+LSTM vs. CfC/LTC at equal imputation (zero-fill / forward-fill / no-imputation); E3 temporal-split check is secondary/exploratory only.

Q11. Does TokaMind (the newer transformer model) already solve this?

A: Not the robustness question. TokaMind outperforms the strongest baseline on clean data (FT-Base beats best baseline on all but Task 4-5, Table 2) and supports optional token/chunk-dropout regularization (TokaMind §5.1), which suggests its authors anticipated the robustness problem – but they never report a systematic test-time corruption benchmark (no NRMSE-vs-corruption-severity table). So even the current best published model for this task has not been evaluated the way we're proposing to evaluate ours.

Q12. Why not reconstruct the complete plasma state (density, temperature, flux together), not just one field?

A: That would be a more complete answer to "state reconstruction," and it's the direction the strongest existing model (TokaMind) is already heading – it is pretrained across 34 different targets jointly. Given our timeline, we scoped this project to a single field (flux map / boundary), which is still a genuine, real reconstruction task with an existing baseline. Multi-field joint reconstruction is noted explicitly as future work rather than attempted here.

Q13. Is flux/boundary reconstruction alone useful, or does real control need more?

A: It is one necessary input, not the complete picture. Real tokamak control systems (e.g. JET's) run on magnetics plus interferometry, polarimetry, spectroscopy, bolometry, and other diagnostics simultaneously, because control decisions need shape/position, temperature, density, radiated power, and disruption precursors together. Our project addresses the shape/position (equilibrium) sub-problem within that larger stack – we do not claim to reconstruct everything a control system needs, and we do not attempt control itself.

Q14. What if the standard CNN+LSTM baseline turns out to already be robust enough?

A: That would still be a valid, reportable finding, not a failed project. The hypothesis that continuous-time encoders degrade more gracefully is motivated by theory and by adjacent evi-

dence (continuous-time models already outperform discrete ones on a different tokamak task), but it is being tested, not assumed. A negative result – no meaningful robustness difference – is scientifically useful and will be reported honestly either way.

Q15. Is your timeline realistic given what's implemented so far?

A: Yes. Literature review and gap finalization are complete. We are currently setting up the TokaMark data pipeline and reproducing the CNN/CNN+LSTM baselines on Task 1-3/1-2, and expect this to be complete within the next one to two days. This keeps us on track against our Work Plan & Timeline: with baseline reproduction closing out on schedule, the remaining work – implementing the CfC/LTC encoder, building the corruption protocol, and running the three planned experiments (E1 clean-data correctness check, E2 main robustness comparison at equal imputation, E3 secondary temporal-split check) – fits within our remaining timeline.

Q16. What hardware / software do you need, and do you have access to it?

A: TokaMark provides existing data loaders and only requires streaming the magnetics/currents/flux signals needed for Task 1-3 (a small subset of the full archive), not the full $\sim$2 TB dataset. Implementation uses PyTorch, with the `ncps` library for CfC/LTC cell implementations, since these are not standard library layers the way LSTM is. We are using free GPU resources – Kaggle Notebooks and Google Colab – for training, which is sufficient given the lightweight input windows and grid sizes involved.

Q17. Why is deep learning appropriate for this problem?

A: Equilibrium reconstruction is traditionally an iterative inverse solve (EFIT on the Grad–Shafranov equation with parameterized profile assumptions), which is too slow for real-time use. The sensor-to-flux-map relation is a high-dimensional nonlinear mapping that can be learned directly from archived shots and evaluated in milliseconds – and TokaMark and TokaMind already show this works well on clean data. Our project does not re-prove that mapping; it tests whether the learned map degrades gracefully when the sensors it depends on are corrupted at test time.